National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Data Science	Course Code:	CS4048
Program:	BS (CS) & BS (SE)	Semester:	Spring 2026
Duration:	60 Minutes	Total Marks:	100
Paper Date:	Feb 25, 2026	Weight	15 %
Section:	All sections	Page(s):	4
Exam Type:	Mid I		

Question	1	2	3			
Marks	/25	/25	/25	/ 25		/100

Q1. (25 Marks) A company wants to understand how two factors—**working hours and sleep hours**—affect employee performance. As an ML engineer, you are asked to build a **multiple linear regression** model to predict employees' performance scores based on these two input features. You are provided with the following data from 3 employees:

X1 working hours	X2 sleep hours	Y performance
1	6	60
2	7	65
3	8	70

In this task, you will create a mo[illegible] [illegible]n:

$$\text{Y-hat} = \theta_0 + \theta_1 x_1 + \theta_2 x_2,$$

where:

- y-hat is the predicted performance score
- x_1 is the number of working hours
- x_2 is the number of sleep hours
- θ_0 is the intercept (bias term)
- θ_1, θ_2 are the coefficients for the input features

Tasks:

1. Estimate the parameters θ_0, θ_1, θ_2 for the multiple linear regression model. Clearly show your calculation steps. You may use any method.
2. Use your trained model to predict the score for a new employee who works 4 hours and sleeps 9 hours.

Equations:

$$X1 = a_{1.23} + b_{12.3} \cdot X2 + b_{13.2} \cdot X3$$

$$\Sigma X1 = N \cdot a_{1.23} + b_{12.3} \cdot \Sigma X2 + b_{13.2} \cdot \Sigma X3 \qquad (1)$$

$$\Sigma X1X2 = a_{1.23} \cdot \Sigma X2 + b_{12.3} \cdot \Sigma (X2)^2 + b_{13.2} \cdot \Sigma X2X3 \qquad (2)$$

$$\Sigma X1X3 = a_{1.23} \cdot \Sigma X3 + b_{12.3} \cdot \Sigma X2X3 + b_{13.2} \cdot \Sigma (X3)^2 \qquad (3)$$

$$x_1 = b_{12.3} \cdot x_2 + b_{13.2} \cdot x_3$$

Where x1 = (X1 – X1 (mean)), x2 = (X2 - X2 (mean)), x3 = (X3 - X3 (mean))

$$b_{12.3} = (\Sigma x_1 x_2)(\Sigma x_3^2) - (\Sigma x_1 x_3)(\Sigma x_2 x_3) / (\Sigma x_2^2)(\Sigma x_3^2) - (\Sigma x_2 x_3)^2 \qquad (4)$$

$$b_{13.2} = (\Sigma x_1 x_3)(\Sigma x_2^2) - (\Sigma x_1 x_2)(\Sigma x_3 x_2) / (\Sigma x_3^2)(\Sigma x_2^2) - (\Sigma x_3 x_2)^2 \qquad (5)$$

Q2. (25 marks) A sports analytics company wants to identify unusual batting performances using statistical techniques. You are provided with the runs scored by 12 cricketers in their most recent match.

Players	Runs
A	44
B	57
C	62
D	49
E	71
F	53
G	60
H	58

I	65
J	73
K	55
L	92

The company defines the following rule:

Any value that does not lie within the **second standard deviation range** is considered an outlier.

Task: Determine the range of the second standard deviation: Identify which player(s), if any, fall outside this range.

Q3. (25 Marks) A local fruit shop owner in Lahore wants to analyze the sales performance of different fruits over the last 10 days. The owner recorded the total daily sales (in kilograms) and now wants to understand:

- Which fruit is selling more
- How sales are distributed
- Whether the sales distribution is symmetric or skewed

Day	Fruits	Sales
1	Mango	120
2	Grapes	85
3	Apples	95
4	Banana	140
5	Orange	110
6	Mango	130
7	Grapes	90

8	Apple	100
9	Banana	150
10	Orange	105

Task: Which type of plot would be most appropriate to visualize the total sales of each fruit? Justify your answer. If you want to analyze the distribution of sales values, which plot would you use and why?
Make appropriate plots

Based on the distribution plot: Determine whether the data is symmetric or skewed. If skewed, specify whether it is: Positively skewed (right skewed), or Negatively skewed (left skewed).

Q4. (25 Marks) A medical researcher wants to study the relationship between **age** and **glucose level** for six subjects. The dataset is as follows:

Subject	Age	Glucose level
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

Calculate the **covariance** between age and glucose level. What does the **sign** of the covariance indicate about their relationship?
Calculate the **correlation coefficient** between age and glucose level. Interpret its value in terms of **strength** and **direction** of the relationship.
Can a **high covariance** always guarantee a strong linear relationship? Explain using this dataset.

Good Luck!